# The Shifting Weight of Nations

A DATA ESSAY, ECONOMIC HISTORY, EIGHTH EDITION

Five centuries of world GDP, the two rulers that make the same country look mighty or modest, and the one variable underneath them both.

An audio edition — figures and tables read aloud with their numbers.

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Inspired by Stephen Kotkin on Hoover's Uncommon Knowledge (host Peter Robinson), youtube.com/watch?v=gBEdxb8ei_0.

Spanning 1500 to 2026, with a 2050 demographic horizon. Eighth (first living edition) edition, June 2026.

Hypothesis. A nation's share of world GDP is not one number but three multiplied together, its share of the world's people, its price level (the ruler you measure with), and its productivity (output per person relative to the world). Strip away the first two and the entire five-century drama of rise, fall, and "re-convergence" collapses onto a single variable, productivity, which is also the only one that will decide the next fifty years.

Approach. Separate the two rulers (market-rate nominal and price-adjusted PPP); add the population-adjusted multiplier that turns a share into a statement about prosperity; triangulate three datasets kept deliberately un-stitched, Maddison for deep history, World Bank WDI for the modern era, the IMF's April 2026 vintage as a present-day marker; weigh every share against the world totals it is a fraction of; decompose the picture by bloc and by who-gained-and-lost; read every economy through all three lenses, country by country; and read every number for direction and order of magnitude, ruler-labelled, never for its decimals.

In one line. Seven conclusions follow, set out at the end. The shortest version: size is headcount; weight is productivity; and the future belongs to whoever holds the frontier of energy and knowledge, not to whoever holds the most people.

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## Totals. The World This Essay Divides

Every share in these pages is a slice of three totals. Here they are first, before any country claims a fraction of them.

People. about 8.2 billion (8.16 billion in 2024 to about 9.66 billion by 2050). headcount, the denominator of the multiplier

World GDP at market exchange rates. $126.3 trillion (current US dollars). nominal, financial command, imports, debt, reserves

World GDP at purchasing power parity. $222.8 trillion (current international dollars). PPP, real output valued at common prices

The two GDP totals are the same world economy seen through the two rulers this essay keeps apart: $126.3 trillion is what global output is worth at market exchange rates; $222.8 trillion is what it amounts to once local price levels are equalised. The roughly $96 trillion between them is the price-level term from the hypothesis, written at planetary scale, the world's "PPP uplift." Population is the third number and the one that converts any share into a statement about how people actually live. Section 1A returns to these totals in full, with the lane distinctions and the denominator warnings; everything in between is the story of how the slices move.

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## Reading. How to Read This Essay

Most confusion in this subject comes from one mistake: letting a single GDP number do three different jobs. A share of world GDP can describe real output, financial command, or per-person productivity, depending on the ruler in use, and most arguments about "who is biggest" are really arguments about which ruler the speaker reached for, usually without saying so. Before any chart, match the question to the instrument.

Who is biggest in real output?. PPP GDP share. It values a haircut in Delhi and one in Zurich at a common price, so it measures volume, not price.. It overstates command over things bought at world prices, chips, jet engines, dollar debt.

Who commands dollars, imports, assets, external finance?. Nominal share. It tracks what output is worth on world markets, imports, collateral, reserves, what you can borrow against.. It lurches with the currency cycle and asset prices, not only with real growth.

Who is actually rich, output per person?. The multiplier (PPP per head). GDP share divided by population share separates size from prosperity.. It is blind to inequality, quality, public goods, and the informal economy.

Why did Japan look so huge in the 1990s?. The FX chapter. The super-yen and asset bubble capitalised Japanese output at a world-market price PPP never endorsed.. Mistaking a currency cycle for economic destiny.

Five centuries of comparison?. Maddison-style PPP benchmarks. Market exchange rates simply do not exist for 1500 or 1700.. The numbers are reconstructions with wide error bars, read shape, not decimals.

Coalition or bloc weight?. Both rulers, summed across members. Alliances need real capacity and financial command at once.. "The bloc" is not a fixed object, the EU, Europe, and the UK are different sets.

The operating rule that runs through every page: read the ruler first, the direction second, and the decimal point last. The direction is usually robust. The decimals, especially before 1900, are estimates wearing the costume of precision.

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## Section 1. The Executive Answer

The story is right. The ruler keeps changing.

A Kotkin-style narrative compresses the world into a few memorable numbers: the United States around a quarter of world GDP with about a twentieth of humanity; Japan from almost eighteen percent to under four; Europe sliding; China returning; India mostly absent. The narrative is useful. It is also dangerous unless every number is labelled with its measuring stick.

The clean revision is this. Market exchange rates measure financial and market weight; PPP measures real output at common prices; the multiplier measures output per person relative to the world average. A country can be enormous because it has many people. It can be rich because each person produces a lot. It can be financially commanding because global markets price its output, currency, securities, institutions, and legal order at a premium. These are related, but they are not the same thing.

The conclusion is not that America is safe forever, that China is destined to dominate, or that Europe is doomed. It is that aggregate share, per-person prosperity, market command, and geopolitical capacity are different layers of power. A serious account must keep them apart long enough to see how they interact.

United States: about 26 percent of world output at market rates, about 15 percent on the PPP ruler, and 4.23 percent of world population. Its output-per-person multiplier is 6.23 at market rates and 3.56 on PPP.

EU-27: about 17 percent of world output at market rates, about 14 percent on the PPP ruler, and 5.59 percent of world population. Its output-per-person multiplier is 3.20 at market rates and 2.57 on PPP.

EU-27 + UK: about 21 percent of world output at market rates, about 16 percent on the PPP ruler, and 6.45 percent of world population. Its output-per-person multiplier is 3.29 at market rates and 2.56 on PPP.

United Kingdom: about 3 percent of world output at market rates, about 2 percent on the PPP ruler, and 0.86 percent of world population. Its output-per-person multiplier is 3.93 at market rates and 2.48 on PPP.

China: about 17 percent of world output at market rates, about 19 percent on the PPP ruler, and 17.51 percent of world population. Its output-per-person multiplier is 0.98 at market rates and 1.13 on PPP.

India: about 3 percent of world output at market rates, about 8 percent on the PPP ruler, and 18.03 percent of world population. Its output-per-person multiplier is 0.20 at market rates and 0.46 on PPP.

Japan: about 3 percent of world output at market rates, about 3 percent on the PPP ruler, and 1.54 percent of world population. Its output-per-person multiplier is 2.40 at market rates and 2.18 on PPP.

Russia: about 1 percent of world output at market rates, about 3 percent on the PPP ruler, and 1.78 percent of world population. Its output-per-person multiplier is 1.10 at market rates and 2.00 on PPP.

World Bank WDI country-sum, 2024. Nominal times is market-weight per person; PPP times is real-output per person relative to the world average. China is huge on PPP but only just above the world average per person. India is huge in people and

potential but still below half the world average per person. The US is smaller than China on PPP aggregate output but much richer and more financially weighty per person.

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## Section 1A. The Totals Behind the Shares

Percentages need denominators; the denominators are world GDP and world population.

A share of world GDP is a fraction. The numerator is a country or bloc; the denominator is the world. Leave the denominator invisible and the reader hears only drama, twenty-five percent, eighteen, four, seventeen. But the denominator itself has changed radically, and it is measured by the same two rulers as everything else.

The market-priced world. In the IMF WEO April 2026 marker, world nominal GDP is about $126.3 trillion in current US dollars. That is the market-exchange-rate world: dollars, imports, external finance, reserve assets, sanctions exposure, and global pricing.

The common-priced world. In the same vintage, world GDP at PPP is about $222.8 trillion in current international dollars. That is the common-price world, real domestic output adjusted for local price levels. It is larger because many goods and services in lower-income economies cost less than their market exchange rates imply.

The headcount. The UN's World Population Prospects 2024 puts world population at about 8.16 billion in 2024, roughly 9.66 billion by 2050, and about 10.18 billion by 2100, peaking near 10.3 billion in the mid-2080s before the first sustained fall of the modern era. That matters because the multiplier is GDP share divided by population share: a country can gain share by adding people, by raising output per person, or by being priced more highly by the ruler in use.

> World nominal GDP. $126.3 trillion. IMF WEO April 2026, 2026 projection, current US dollars. Market-exchange-rate denominator for financial command, imports, dollar debt, valuation, sanctions.

World GDP at PPP. $222.8 trillion. IMF WEO April 2026, 2026 projection, current international dollars. Real-output denominator at common prices. Not spendable FX; a statistical purchasing-power dollar.

World population. 8.16 billion (2024) to about 9.66 billion (2050) to peak about 10.3 billion (2080s). UN WPP 2024, medium variant. The denominator for the multiplier. Growth slows, but the world still adds about 1.5 billion people by 2050.

Report internal lane. ~$110 trillion nominal; ~$170 trillion PPP (constant 2021 int\$, about $195 trillion current); 8.05 billion, 2024. WDI/QoG country-sum. Used for the internal share and multiplier arithmetic, against a single common denominator.

Denominator warning. This report carries two modern denominators and keeps them apart by design. Official IMF/World Bank world aggregates are best for a current global total; the WDI/QoG country-sum denominator is best for the internal share arithmetic, because every entity is then measured against the same source lane. The two are close enough for the argument but not identical, a few tenths of a percentage point apart. A third lane, the WDI constant 2021 international-dollar series (~$170 trillion for 2024), values real output at a fixed benchmark and is lower again. These three PPP totals, $222.8 trillion (2026 current), ~$195 trillion (2024 current), ~$170 trillion (2024 constant-2021), answer different questions and are never spliced into one line.

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## Section 2. One Identity, Three Lenses

Before the curves, the algebra, because it is the whole argument in one line.

Any country's share of world output can be written as the product of three ratios:

The spine of the essay is one identity. A country's share of world GDP equals its share of world population, times the price level, the ruler you measure with, times its productivity, meaning output per person relative to the world. Population is why old Asia looked huge; the price level is the whole nominal-versus-PPP gap; productivity is what the essay is about.

Read left to right, this is why the charts that follow disagree with one another. Population explains why China and India were "huge" in 1700 and why they are rising again now. The price level is the ruler, the entire gap between the nominal and PPP pictures, and it moves with currencies and inflation as much as with real output. Productivity is what is left when you divide the first two out: output per person, relative to the world average. That last term is what this essay is really about. It is what the Great Divergence created, what the Asian re-convergence has so far barely touched, and what the technologies of the next half-century will redistribute.

This report therefore refuses a single master line. It uses three lenses and keeps them labelled.

> Nominal GDP share. What is the economy worth at market exchange rates?. Imports, debt, reserves, traded goods, financial command, sanctions, valuation.. Treating currency swings as pure real growth or decline.
>
> PPP GDP share. How much real output is produced at common prices?. Cross-country real output, living standards, centuries-long history.. Pretending PPP buys imported chips, jet engines, oil, or dollar debt at local prices.
>
> Multiplier. Output per person relative to the world average?. Separating size from prosperity, population from productivity.. Ignoring inequality, quality, public goods, and strategic conversion.

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## Section 3. PPP Explained Slowly

PPP is necessary. PPP is insufficient. Both statements matter.

Purchasing power parity begins with a simple observation: prices differ across places. A dollar converted at the market rate buys a different bundle in Delhi, Tokyo, Istanbul, Zurich, or Kansas. Non-traded services are especially different, a haircut, rent, local transport, a bowl of rice can cost far less in a lower-income country. Compare GDP only at market exchange rates and you may badly understate the real volume of domestic goods and services in lower-price economies. PPP corrects that by asking what GDP would look like priced with a common international structure rather than market rates. That is why China and India become much larger on PPP

charts than on nominal ones, and why PPP is the right ruler for real output, material well-being, and long-run history.

What PPP gets right. It values output at common prices, so low-price economies are not mechanically shrunk by weak currencies. It is closer to domestic command over local goods and services than nominal dollars are. It is the only plausible ruler before modern exchange-rate markets existed. And it lets us see when a populous country is actually producing large real output rather than merely receiving a low dollar price.

Where PPP wobbles, and it does. It is not a currency you can spend abroad, not a reserve asset, and it does not mean that Chinese, Indian, Russian, or Turkish income can buy semiconductors, aircraft engines, oil, LNG cargoes, medical devices, dollar bonds, or naval hardware at local service-sector prices. The PPP dollar is a statistical dollar, not a shipping invoice. It also depends on price surveys, national accounts, expenditure weights, imputation, and benchmark years. The International Comparison Program behind it runs only every six years or so, covers thousands of goods across some 180 countries, and is genuinely hard: the 2005 round cut China's and India's estimated real incomes by roughly 40 percent overnight; the 2011 round then raised China's by about 20 percent. Current PPP figures carry something like ±5 to 10 percent structural uncertainty, materially more than directly reported nominal figures, and the famous 2014 "China overtakes the US" story was, strictly, premature.

Imports and traded goods. PPP overstates command over goods bought at world prices. Use nominal or external-balance measures.

Debt and reserves. Dollar debt and reserves are nominal claims. PPP does not pay them.

Military procurement. Domestic labour may be PPP-relevant; imported chips, engines, sensors, and machine tools are nominal/traded.

Quality and inequality. PPP averages do not guarantee equivalent quality, distribution, public goods, or household welfare.

Revisions. ICP rounds move countries by tens of percent; historical reconstructions have wide error bands.

Exchange-rate forecasting. PPP says nothing about where a currency should trade tomorrow; never use it as a standalone valuation signal.

PPP in one sentence: it is the best ruler for real output at common prices, nominal is the better ruler for world-market command, and the multiplier turns either into a per-person punch. None of the three should be used alone.

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## Section 4. Deep History, The Contested Origins

The long arc is a population story until it becomes a technology story.

`Figure 01.` Five centuries of relative economic size. PPP-style world GDP shares at benchmark years, 1500 to 2024, shown as benchmark bars (honest about where data exists) so the agrarian-Asia to Western-surge to Asian-re-convergence rotation is visible without inventing decades nobody measured.

`Figure 02.` The same five centuries as a re-sliced pie. The cumulative 100 percent view; seductive precisely because it looks continuous, which is why Figure 01's honest gaps sit beside it.

Before industrialisation, aggregate economic size was heavily tied to population, land, calories, and labour. China and India were huge because they held large shares of humanity organised around intensive agriculture, urban craft, and long commercial traditions. That size should not be confused with rich-country living standards. A large preindustrial share often meant many people producing near-subsistence output, not a high multiplier. China's third of the world in 1820 was not a story of rich Chinese; it was a story of many Chinese, at roughly subsistence income, in a world where almost everyone lived at subsistence. Aggregate size in the pre-industrial era was very nearly a headcount.

These deep-history numbers are estimates with wide error bars, and the scholarship is alive and quarrelsome. Angus Maddison's figures are the backbone of nearly every chart you will ever see, including these. But a school led by Stephen Broadberry, rebuilding national accounts from wages, prices, and tax records, has steadily revised the picture, and the revisions cut against the romance of a rich East suddenly overtaken. On their reconstructions the divergence in living standards between northwest Europe and the leading Asian economies began far earlier than

the canonical "Great Divergence around 1800" of Kenneth Pomeranz and the California school: comparing the largest Asian state with the leading European ones, the gap is already visible by around 1400. China's Ming-era output per head has been marked down by up to a third; India looks prosperous at the Mughal peak around 1600, then drains away through the eighteenth century, before the British Raj proper even begins. So treat the headline "China about 40 percent" as a rounded high-end ceiling, Maddison's own peak for China is closer to a third (about 33 percent in 1820), and even the textbook moment when the United States overtook Britain in output per head has been pushed back to the 1870s.

If the divergence is real, what caused it? Three explanations compete, and the honest answer is that the field has not settled, so neither will this essay. The first is ecological: Pomeranz argues Britain happened to sit on accessible coal and to command New-World land that relieved the Malthusian ceiling, so its escape was as much luck and geography as genius. The second is institutional: in the tradition of Douglass North and, more recently, Acemoglu and Robinson, secure property, open inquiry, patentable return, and constraints on the sovereign let Europe convert invention into compounding growth. The third is the one the numbers keep gesturing at, and that a cyclical historian like Ibn Khaldun would press: institutional stagnation inside the great Asian empires, an agrarian-bureaucratic equilibrium that taxed the surplus without reinvesting it, so that Song China's medieval efflorescence never compounded into take-off. Joseph Needham's famous puzzle, why the civilisation that invented paper, printing, gunpowder, and the compass did not have the Scientific Revolution, lives in this third box.

And there is a fourth factor the curves are unusually good at hiding, because a falling line names no culprit: power. India's share did not simply fail to rise; large parts of it were actively de-industrialised under a specific colonial relationship, a hollowing-out of its textile economy by a mix of British factory competition, the collapse of Mughal demand, and the policy architecture of conquest. The next chart makes that violence visible, because it measures the one thing GDP share blurs: who actually made the world's goods.

## Section 5. Manufacturing, Not GDP

GDP share and industrial command are related, but not identical, and the manufacturing ruler moves far more violently.

`Figure 03.` Share of world manufacturing output (Bairoch), 1750 to 1900. The most violent reversal in the whole story, and where the "China was a third of the world" figure actually comes from.

Modern power did not come from GDP volume alone. It came from a particular kind of output: mechanised, scalable, energy-intensive production linked to finance, logistics, weaponry, and science. A country could stay large in agrarian output while losing strategic centrality if manufacturing shifted elsewhere.

In 1750, China (32.8 percent) and India (24.5 percent) together made about 57 percent of the world's manufactures, this, not GDP, is the source of the "Asia was a third of the world" figure people half-remember. Over the next 150 years it collapses: India to 1.7 percent by 1900, China to 6.2 percent. Meanwhile Europe as a whole climbs from 23 percent to 62 percent, the UK alone going from under 2 percent to nearly 23 percent at its 1880 peak, and the United States rockets from a rounding error to 24 percent. The non-Western share fell from 73 percent to 11 percent. The subtle part: even in 1750, when Britain made under 2 percent of world manufactures, its output per person already ran ahead of India's and China's. The non-Western collapse was not merely relative, India's and China's manufacturing output per head actually fell in absolute terms while Europe's multiplied roughly eightfold.

Two lessons travel out of this chart. First, "share of manufacturing" and "share of GDP" are different rulers again, and the manufacturing one moves far more violently, which is exactly why the round "a third / 40 percent" number for old Asia feels so large: it is frequently the manufacturing figure smuggled into a GDP sentence. Second, and pointing forward to the thesis: what the West acquired in the nineteenth century was not a bigger population but a new way of converting energy into goods, coal and steam doing the work of millions of hands. That is the first appearance of the variable everything else reduces to.

## Section 6. The Market-Weight Era, Kotkin's Actual Canvas

The same years, two different rulers; and most of his modern figures live on the nominal one.

`Figure 04.` World GDP share, nominal (market exchange rates), 1960 to 2024, current US dollars. Where "Japan 18 percent," "US 25 percent," and "Europe 30 to 17" live. Watch how much the lines move with currencies, not just with growth.

The modern period is where the soundbites live and where ruler confusion does the most damage. On market exchange rates the US runs from a postwar about 40 percent (1960) down toward a long-run about 25 percent, jaggedly, because exchange rates whip the figure around. Japan's 1995 spike to about 18 percent is the super-yen and the asset bubble, not real output doubling; the "lost decades" and a strong dollar then more than halve it. Europe's path wobbles with the euro. China is the one secular climb, under 2 percent in 1990, a peak near 18 percent in 2021, easing since. India barely moves on this ruler until very recently. Toggle the West bloc and the headline appears: the US-plus-allies share fell from about three-quarters to roughly half.

`Figure 05.` The American century in one line: US nominal share, 1913 to 2024. The about 50 percent "rubble peak" of 1945, bracketed by about 19 percent in 1913 and about 26 percent today.

It pays to be precise about why a nominal line lurches, because "currency" is a lazy shorthand. A country's nominal share can move for four quite different reasons, and only the first is what most people think they are reading. One: real growth, it actually makes more stuff. Two: relative inflation, if its prices rise faster than the world's, its nominal GDP swells even with flat real output. Three: the exchange-rate cycle, when the dollar strengthens, every other country's dollar-denominated GDP shrinks mechanically, and the US share rises without a single extra widget. Four: policy, a managed currency, like the yuan, can be held down to protect exporters, suppressing the country's own measured nominal share. Japan's mountain and China's recent dip are both mostly reasons two and three, not one. Hold that four-way split; the dollar section makes it load-bearing.

## Section 7. The Multiplier, Strip Out Population

This is the population-adjusted punch, and it is the chart that breaks the myth.

Figure 06. Income relative to the world average, PPP, 1820 to 2024. GDP share ÷ population share; the dashed line is the world mean (1.0).

The multiplier is the most intuitive and most neglected measure in the conversation. It is simply GDP share divided by population share. A country with 25 percent of world GDP and 5 percent of world population produces about five times the world-average output per person. A multiplier of 1.0 is the world's per-capita mean; 3.0 means three times as rich as the typical human.

The multiplier turns size into prosperity, and it turns the rise of Asia into a more precise story. China's "mighty" 33 percent of 1820 was a multiplier of just 0.9, its people sat at, even slightly below, the world average. It then crashed to 0.2 across the century of humiliation and the Mao era, and has only just clawed back to about 1.1. India is at about 0.46: real progress, still under half the world average, a stage behind China. Meanwhile the US (about 3.6 times ) and Western Europe (about 2.7 times ) have sat three to four times above the mean for over a century, and Japan's leap from about 0.9 to about 3 is the postwar miracle in a single line.

> *The rise of Asia in size is largely population weight returning. The rise of Asia in prosperity has barely begun.*

It is worth pausing on what a multiplier of 0.46 means, because the decimal hides a person. India's figure is a distance: between a life in Munich and a life in Bihar, between a clinic with an MRI and a clinic with a queue, multiplied across one and a half billion people. China's climb from 0.2 to 1.1 is the largest movement of human beings out of poverty in recorded history, and it still only reaches the world's middle. Every share point in every chart here contains hundreds of millions of such lives; the multiplier is the closest any of these numbers comes to telling you how those lives are lived. That is the difference between counting people and weighing what their days contain.

## Section 8. The Same Year, Measured Four Ways

Drop the five-century arc and hold a single year still.

Figure 07. Nominal vs PPP share of world GDP, 2024. Darker bar = market exchange rates, lighter = PPP.

Switching from market rates to PPP lowers the share of high-price economies, the US (26 to 15), the EU (18 to 14), Japan, the UK, and raises the share of lower-price, populous ones, China (17 to 20, overtaking the US), India (3.6 to 8.4), Russia. This is why Kotkin's "Europe about 17 percent" matches the PPP figure for Europe while his "Japan 18 percent" and "US 25 percent" use the nominal one. Both columns are correct; they answer different questions, financial weight on the dark bars, real output on the light.

Figure 08. The 2024 ruler gap. Nominal share minus PPP share, percentage points; positive = nominal-heavy, negative = PPP-lifted.

Stack those two columns into a single bar per country and the gap becomes a number in its own right, the cleanest one-glance map of who the dollar flatters and whom PPP lifts. Positive bars are economies that look bigger in dollars than in real volume: the United States most of all (+11.3 points), where high domestic prices and the reserve-currency premium inflate the dollar value of output; then Europe and the UK. Negative bars are economies PPP lifts: India (−4.8), China (−2.5), and Russia. The size of each bar is, in effect, the price-level term from the opening identity, read off one economy at a time. It is also a preview of the convergence clock: a converging country's positive gap grows as its prices rise toward the frontier's.

Figure 09. Income vs world average, 2024, nominal vs PPP. The fulcrum is China.

If the ruler changed only the totals, every pair would scale together. It doesn't, and one country flips sides. On the nominal ruler the US looks like a 6.2 times colossus; on PPP it is a still-formidable but humbler 3.6 times. The UK and Russia move most in proportion. But the verdict for almost every economy survives the switch, except China, at 0.98 times nominal and 1.13 times PPP. On one ruler the average Chinese citizen is just below the world mean; on the other, just above. The single most consequential country on earth straddles the parity line, and which side it lands on

is a measurement choice. India, at 0.20 times nominal and 0.46 times PPP, is unambiguously below on either.

Figure 10. Population weight vs real-output weight, 2024. The multiplier, drawn as a map: the diagonal is parity, and the vertical distance from it is the multiplier.

The United States, Europe, Japan and the UK float high above the line on small populations, the frontier economies. China sits almost exactly on it, the giant that has reached the middle. India sits far below and to the right: more than a sixth of humanity, still well under the parity line. The whole essay is an argument about how points move toward that line, or away.

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## Section 9. The FX Chapter, Japan and the Price-Level Clock

Currency cycles can masquerade as economic destiny.

Figure 11. Japan, the warning label: one country, two rulers, 1990 to 2024. Solid = nominal, dashed = PPP.

The Japan claim is the cleanest warning label in the whole debate. Its nominal share peaks at about 18 percent in 1994, the super-yen and the asset bubble capitalising Japanese output at a world-market price, then more than halves as the bubble deflates and the dollar strengthens. The PPP line never crosses about 8 percent and slides quietly to about 3.4 percent today. The gap between the two is the currency-and-price-level premium, not a doubling of real output. Japan is the cleanest proof in the essay that a number can halve on one ruler while the thing it claims to measure, real output per person, barely moves.

The right revision is to refuse the habit of treating nominal GDP as a sloppy measure. Nominal GDP is the correct measure for certain questions: imports, external debt, reserves, dollar finance, market acquisitions, sanctions, and the price paid for traded goods. It is just the wrong measure for others: domestic real output and average material well-being.

> Real growth. More output can raise nominal share.
>
> Relative inflation. Higher domestic prices can lift dollar GDP without equivalent volume gain.

Exchange rates. A strong currency can inflate share; depreciation can hide real volume.

Asset / capital-market valuation. Bubbles and risk premia can enter the national market-weight story.

Capital controls / sanctions. A country can have PPP volume but weak external convertibility.

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## Section 10. Geopolitics (I), Who Really Paid for China's Rise

Start with the peak everyone cites, the most misleading true sentence in the whole debate.

In 1945 the United States held something like 50 percent of global output in nominal terms, plus 80 percent of the world's hard-currency reserves and over half its manufacturing. But that share measured the rest of the world lying in rubble as much as American strength: Germany, Japan, Britain, France and the USSR had had their industrial bases bombed, blockaded, or bled. As they rebuilt, the US share had to fall even as American output kept growing. By 1960 it was about 40 percent nominal; on the price-adjusted PPP ruler, the US "high noon" around 1950 was already only about 27 percent. Same country, same era, 50, 40, or 27 percent depending entirely on the year and the ruler. Any single headline number here is a choice, not a fact.

`Figure 12.` US-plus-allies vs China-plus-India, both rulers, 1990 to 2024. WDI country-sum; solid = nominal, dashed = PPP.

`Figure 13.` Who gained and lost world-GDP share, 1990 to 2024. Change in percentage points; darker = nominal, lighter = PPP.

On the nominal ruler the industrial-allied bloc, US + EU + UK + Japan, falls from about 76 percent to 51 percent of world GDP since 1990, while China + India climb from about 3 percent to about 21 percent. On PPP the bloc falls further (59 percent to 35 percent) and China+India rise higher (7 percent to 28 percent), because PPP reweights lower-price economies up. Either way the bloc lost roughly a quarter of the world. But the decomposition overturns the lazy version of "Western decline." China gained about 16 points on both rulers. The losers are Europe (EU+UK, −13

points on both) and Japan (–11 nominal). The United States is nearly flat in nominal terms (–1.1), it held its quarter of the world for four decades, and only lower on PPP (–6.5) because everyone's PPP slice was diluted by the populous risers. The headline "the West is in relative decline" is accurate; the sub-headline "and the United States is the part that declined" is not. America held its ground; its allies ceded theirs.

That asymmetry, patron stable, clients shrinking, is the geopolitically loaded fact under all the arithmetic, and it cuts two ways. A patron whose allies shrink gains in relative standing but loses the very coalition that converts standing into power: alliances are a force-multiplier a bare GDP share does not show. The same numbers that comfort Washington should also unsettle it.

GDP share is not geopolitical power by itself; it is a capacity measure that must be converted. Conversion requires fiscal extraction, institutions, military procurement, logistics, alliances, energy, technology, standards, and political legitimacy. A large economy with weak conversion can underperform strategically; a smaller economy inside a strong alliance and dollar system can overperform.

PPP domestic volume. Local labour, services, construction, domestic military inputs.. Quality gaps, bottlenecks, corruption, weak logistics, import dependence.

Nominal dollar output. Imports, foreign debt service, reserves, market acquisitions.. Currency depreciation, sanctions, capital controls, dollar funding stress.

High multiplier. Tax capacity, research systems, complex services, advanced firms.. Inequality, political fragmentation, aging, fiscal rigidity.

Large population. Scale, labour force, domestic market, military manpower.. Low human capital, weak job creation, energy scarcity, poor institutions.

Alliance network. Shared technology, basing, standards, financial systems, defence pooling.. Coordination failures, free-riding, divergent threat perceptions.

## Section 11. Geopolitics (II), The Ruler Is a Weapon

Which ruler you choose is itself a claim about the world, and the great powers choose accordingly.

By now the two-rulers point should feel less like methodology and more like contested ground, because it is. Beijing prefers PPP: on that ruler China is already the world's largest economy, the "East is rising" narrative writes itself, and the implied demand for a larger voice in the institutions of the world order follows. Washington prefers nominal: on that ruler the United States still leads, the dollar system is the proof, and "America is not in decline, its allies are" is the natural reading. Neither is lying. Each has simply selected the instrument that frames the contest in its favour. A reader who can spot which ruler a pundit reached for can usually infer the argument before it is made.

The dollar is the hidden denominator. There is a deeper asymmetry buried in the nominal ruler, the mirror image of the PPP point about China. Every other country's nominal share is measured in the unit the United States prints. That confers a quiet flattery, reserve-currency demand and high domestic prices inflate the measured dollar value of US output, and, more importantly, a structural option no one else holds: when a global crisis hits, capital flees to the dollar, so the US share tends to rise exactly when the world is most frightened. The euro has no such put; the yuan, with a closed capital account, is nowhere near acquiring one. A structurally weaker dollar, from twin deficits, debt monetisation, or a serious move away from dollar reserves, would lower the US nominal share and lift everyone else's without a single real change in output, just as a strong dollar has flattered it. But the data counsel patience with the de-dollarisation story: the dollar is still around 58 percent of global reserves, no rival is close, and the yuan sits below 3 percent. It is a slow tail risk, not a base case, the reserve term decays in decades, not years.

It is worth being blunt about what the nominal ruler is and is not, because the temptation is to treat it as the vain twin of "real" PPP. It isn't. The desk that trades these numbers keeps having to make the same four corrections.

Nominal strength isn't vanity.. Dollar-priced output is what buys imports, services foreign debt, anchors reserves, and funds the institutions that turn weight into the ability to act.

Currency weakness isn't mere optics.. A softer yen, euro, or yuan really does shrink a country's dollar share and its command over traded goods, even with flat real output at home.

PPP is not an exchange-rate forecast.. It says nothing about where a currency should trade tomorrow, and should never be a standalone valuation signal.

Sanctions and capital controls split the two rulers.. An economy can hold real domestic capacity on PPP and still be locked out of the foreign technology, finance, and settlement systems only nominal dollars reach. The 2022 reserve freeze is the worked example: Russia's PPP output was untouched; its access to that output's dollar value was not.

"Friend-shoring," trade blocs, and the weaponisation of finance, sanctions, the 2022 reserve freeze, are all attempts to manage the apparatus that converts standing into power, and all of them, pushed far enough, fragment the single world economy whose "shares" this essay has been measuring. A fragmented world does not have one GDP pie to slice; it has two or three, measured in rival units. That is the scenario the futures section prices.

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## Section 12. The Convergence Clock

Does the gap between the two rulers stay fixed? It does not, and the way it moves is the most useful forward-looking idea here.

The mechanism has a name, the Balassa, Samuelson effect, and the intuition is simple. Poor, fast-growing economies have cheap non-tradable services, the haircut, the bus ride, the meal, so their PPP output looks much larger than their nominal output (China: 19.7 percent PPP versus 17.2 percent nominal; India: 8.4 percent versus 3.6 percent). As they get richer, productivity in their tradable sectors rises, wages rise across the whole economy, and the price level rises with them, so their nominal share climbs toward their PPP share. The gap between the two rulers, in other words, is a clock: it measures how far a country still has to travel.

This reframes the essay's ending. For a converging economy the gap closes from below as prosperity rises, China's nominal share will drift up toward its PPP share over the coming decades even on unchanged real growth, simply from price-level convergence; India's gap, wider still, is a longer clock with more to run. But the clock can also stall: if productivity growth stops, the middle-income trap, the fate that caught much of Latin America and arguably now Italy, the price level stops rising and the nominal share plateaus permanently below the PPP share. So the gap is both a promise and a warning. China's fork is exactly here: glide toward the frontier, or stall at the institutional ceiling a cyclical historian like Ibn Khaldun would say has caught every over-extended bureaucratic empire before it.

Figure 14. Why India is the swing variable. Implied PPP share of world GDP if 2024 population is held fixed and the multiplier is swept; dots mark each economy's 2024 position.

Two structural forces set the pace of that clock. The first is demography, and here the contrast is stark. China, Europe, Japan, and South Korea are aging and beginning to shrink, a falling working-age share is a headwind on both growth and the multiplier. India has a young and still-growing workforce: its demographic dividend is the wind at its back, which is why even modest per-head gains, multiplied across 18 percent of humanity, move the whole world pie. Each line in Figure 14 holds an economy's population share fixed and sweeps only its multiplier, so the steepness of the line is the population base. India and China have far the steepest slopes, roughly a sixth of humanity each, so a one-point gain in the multiplier adds about 18 points of world share for either, against barely 4 for the United States on a twentieth of the people. What makes India specifically the swing variable is where it starts: at 0.46 times it has the most room to climb, and a young workforce to climb with, while China at 1.13 times has already reached the middle and is aging. Lift India from 0.46 times to merely the world average and its share roughly doubles, from about 8 percent toward about 18 percent, the single largest latent move on the board.

The second force is the one this essay, like Kotkin's omission of India, has itself under-weighted until now: Africa and the rest of the world. The grey "Rest of World" band in Figure 02 is already over a third of global output and the fastest-growing segment; sub-Saharan Africa will supply most of the planet's population growth this

century, and by 2050 roughly one in four people on earth will be African. The next "missing giant," the India of the 2050s, is being born there now. An essay that measured the weight of nations and stopped at the seven Kotkin named would repeat the very omission it set out to correct.

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## Section 13. Technology, The Engine Under the Numbers

Return to the variable everything reduced to: productivity, the third term in the identity.

The deep history of productivity is, at bottom, the history of one thing: how much energy a society can harness through knowledge and convert into useful work per pair of hands. That is the thread tying Bairoch's eighteenth-century weaver to the data centre. Britain's leap was not more weavers; it was coal and steam doing the work of millions of them. The multiplier the United States has held for 150 years is not American virtue; it is the compounding of successive general-purpose technologies, each a new way of turning energy and information into output.

Read the long arc as a relay of these technologies. Steam (c. 1780 to 1840) broke the link between muscle and output and gave Britain its manufacturing surge. Electricity and the internal-combustion engine (c. 1880 to 1940) handed the frontier to the United States and Germany, precisely when the US multiplier vaults above 4. Information and communications technology (c. 1970 to 2010) extended the American lead into software and finance and helped open the door for the East-Asian export model. Each wave widened the gap between the societies at the frontier and everyone else; the Great Divergence is, in this light, simply the first time one civilisation pulled decisively ahead on energy-per-person and stayed there. China's recent climb is the mirror image: the fastest adoption in history of technologies the frontier had already built.

> Steam, coal, factories. Broke the agrarian population-output link and drove the Western surge.
>
> Railways, electricity, mass production. Created national markets, scale economies, and high industrial multipliers.

Computers, internet, platforms. Helped the US retain nominal and productivity weight while Europe and Japan lost relative share.

Advanced manufacturing and supply chains. Enabled China to convert population into industrial scale with exceptional speed.

AI, chips, data centres, robotics. Could re-concentrate frontier productivity or diffuse multiplier gains.

Energy transition and climate adaptation. Will reshape costs, industrial locations, mineral politics, and infrastructure productivity.

Which sets up the live question. The next general-purpose technology, artificial intelligence, layered on cheap electricity, could push the frontier in either of two opposite directions, and the honest position is that we do not yet know which. If AI mostly raises the frontier, compounding fastest where capital, data, and elite talent already cluster, it will widen the US and Chinese multipliers and re-diverge the world. If instead it diffuses fast and cheap, a tutor, a doctor, an engineer in every phone, it becomes the greatest catch-up accelerant ever built, lifting India's and Africa's multipliers toward the mean. The same technology, two vectors. Whichever wins will move these charts more than any demographic trend, because it acts directly on the only term that matters. Technology is not a miracle word: it changes GDP share only when it changes output per person at scale. Chips without electricity, factories without logistics, or AI without institutions do not automatically become national power.

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## Section 14. The Future, Three Futures, and What to Watch

No honest version of this subject ends in a forecast line.

The distribution of future shares is fat-tailed and asymmetric: the modal path is slow convergence, but the tails, a dollar crisis, a yuan float, a war over Taiwan, an AI-productivity discontinuity, are violent and would dominate everything. So the essay ends not with a prediction but with a cone: three regimes, each internally coherent, with the signals that would tell you which one you are living in.

Convergence, the modal path. China's nominal share drifts up toward its PPP share as its price level rises; India's wider gap slowly narrows; the US nominal share

eases toward the low-20s as the dollar normalises; the West bloc keeps gently sliding. The world of 2050 is multipolar but still integrated, one pie, more evenly cut. Watch: yuan policy; US inflation versus trading partners; India's GDP-per-head breaking above ~$5,000.

Fragmentation, the bloc path. Trade blocs harden, friend-shoring deepens, reserves diversify, and the single world economy splits into rival currency-and-supply spheres. "World GDP share" becomes a contested statistic measured in different units, the question itself fractures. Watch: tariff and export-control regimes; the dollar's reserve share; the build-out of non-dollar payment rails.

Frontier discontinuity, the fat tail. An AI-plus-energy productivity jump concentrates gains where capital, compute, and talent already cluster, the US and China, widening multipliers and re-diverging a world that had been converging. Size matters less than ever; the frontier matters more. Watch: productivity-growth data; compute and electricity build-out; whether AI gains diffuse or concentrate.

All three turn on the same hinge, productivity, and specifically who commands the energy-and-knowledge frontier. That is the thesis returning as a forecast: population set the stage and the ruler sets the framing, but it is the third term, productivity, that decides which of these worlds we wake up in.

Demography. Young workforces in India and the Global South become productive labour.. Aging shrinks the working-age share in China, Europe, Japan, Russia.. India, China, EU, Japan

Technology & AI. Gains diffuse, a tutor, a clinician, an engineer in every phone.. Gains stay narrow; valuations correct; compute and power bottlenecks bind.. US, China, India

Energy. Cheap, clean electricity lowers the cost of making everything.. Price shocks, climate damage, mineral chokepoints raise costs and split supply chains.. EU, China, India

Institutions. States discipline incumbents, reward merit, build human capital.. Middle-income traps, elite capture, polarisation stall convergence.. China, India, US

The dollar cycle. Stable currencies and deep markets support nominal share and import power.. Strong-dollar stress, debt strain, or fragmentation prise nominal apart from PPP.. Emerging markets, euro area, Japan

The three middle rows act directly on productivity; the two outer ones set the headcount and move the nominal ruler. That is why the middle decides more than the edges.

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## Section 15. The Actors, One by One

Six readings off the same two rulers, each read through share, ruler, and multiplier.

United States, 26.3 percent nominal, 15.1 percent PPP, 4.2 percent of people, times 6.2 / times 3.6. The anomaly the whole essay keeps circling. America holds about a quarter of world output at market rates on roughly one twenty-fifth of its people, and it is the one rich economy that did not cede ground as China rose. None of that is a population story; it is the compounding of a century and a half at the energy-and-knowledge frontier, plus the quiet flattery of pricing the world in the unit you print. Its forward question is whether the next general-purpose technology keeps lifting US productivity, or whether fiscal strain and a narrowing of the gains hollow the multiplier out while the dollar still says otherwise.

China, 17.2 percent nominal, 19.7 percent PPP, 17.5 percent of people, times 0.98 / times 1.13. The only major economy that changes sides of the parity line depending on the ruler. Its PPP output is now the largest on earth, and its climb from a 0.2 multiplier to 1.1 is the largest movement of people toward the world's middle in recorded history; but 1.1 is the middle, not the frontier. China has done the hard part of scale, industrial depth, infrastructure, a market large enough to set world prices, and now meets the part the curves have always punished: turning catch-up into frontier productivity before aging, debt, and a closed capital account close the window.

India, 3.6 percent nominal, 8.4 percent PPP, 18.0 percent of people, times 0.20 / times 0.46. Not a small economy, a very large one with a low floor. India already holds more than a sixth of humanity and more than a twelfth of world output on

PPP, yet sits below half the world average per person, a clear stage behind China on every ruler. That gap is precisely why it is the wildcard: with a base this large, even a modest rise in output per head moves the whole world pie, and its young workforce is the demographic wind no aging giant has. Whether that potential becomes weight turns on the unglamorous machinery of productivity, human capital, reliable power, female labour-force participation, manufacturing depth, not on the headcount it already holds.

Europe + the UK, 21.3 percent nominal, 16.5 percent PPP, 6.5 percent of people, times 3.2 / times 2.6. Rich, not poor, and smaller in relative share than the nominal charts of the 1990s suggest. The bloc still runs a PPP multiplier above 2.5, with deep human capital, advanced manufacturing, and regulatory reach; what it lacks is the population weight of a continental rival and the frontier-scale technology firms that let the United States convert size into platforms. Most of the West's lost share was paid here, not in Washington. Its forward question is whether an aging, fragmented bloc can organise its latent capacity, in defence, energy, and digital, into renewed productivity, or settle into a dignified plateau.

Japan, 3.7 percent nominal, 3.4 percent PPP, 1.5 percent of people, times 2.4 / times 2.2. The warning label of the whole debate, and it should be taught as one. The famous fall "from 18 percent to 4 percent" is almost entirely a nominal-ruler story, the 1990s super-yen and asset bubble unwinding, plus slow growth, and Japan's PPP share never resembled that mountain. Strip the currency away and what remains is an advanced, high-multiplier economy that stopped growing its share, not a collapse.

Russia, 2.0 percent nominal, 3.6 percent PPP, 1.8 percent of people, times 1.10 / times 2.00. The reminder that GDP share bounds power without defining it. Russia looks markedly larger on PPP than nominal, cheap domestic prices nearly double its multiplier, which is the right ruler for soldiers, fuel, and steel paid for at home, and the wrong one for imported chips, jet engines, and dollar finance, which arrive at world prices a sanctioned economy cannot dodge. Its weight on the page is small; its capacity to impose costs, through energy, nuclear arms, and territorial depth, is not. It is the case where the two rulers split not by accident but by policy.

## Section 16. Stepping Back, The Size of the Whole

Every share in this essay is a slice of something; before the last word, weigh the whole.

Figure 15. Five centuries of the whole: headcount and real output, 1500 to 2000. Constant 2011 international dollars, PPP; deliberately not spliced to the current-dollar totals.

Figure 16. The modern totals, and the road to 2050. World population 1900 to 2024 with the UN's median projection; world GDP at current prices, nominal and PPP. Current-dollar terms, a different lane from Figure 15.

World output in 2024 was about $110 trillion at market exchange rates, and roughly $195 trillion at purchasing-power parity in current international dollars, the same two rulers, now laid against the planet. About 8.2 billion people produced it. On the IMF's April 2026 path both totals rise further, toward roughly $126 trillion nominal and $223 trillion PPP for 2026. Population climbed from about 1.65 billion in 1900 to 8.2 billion in 2024 and is projected near 9.7 billion by 2050, peaking around 10.3 billion in the 2080s before turning down. Most of the growth still to come is African, the next missing giant, being born where the essay's demography pointed.

Hold the two pictures together and the claim is visible in one sweep. Across five centuries the headcount rose roughly fourteenfold; real output rose more than a hundredfold. Output ran far ahead of population, and the widening gap between the bars and the line is output per person, the productivity term from the identity, almost all of it earned in the last two centuries. The shares are how the pie is cut; the totals are how much there is to cut, and how fast it is still growing.

## Section 17. Claim Audit

A useful soundbite is a ruler-labelled soundbite. Most of Kotkin's claims are directionally right; they go wrong only when the ruler is left unnamed and a PPP fact is read as a nominal one, or vice versa.

US about 25 percent of world GDP since about 1880. Nominal. Holds. Long-run average near a quarter; about 26 percent nominal in 2024, about 15 percent on PPP. The 2011 trough was about 21 to 23 percent, and the line has recovered since.

Postwar peak about 50 percent was the anomaly. Both. Holds, and is more ruler-dependent than it sounds. about 50 percent nominal in 1945 (a rubble artifact), about 40 percent by 1960, but only about 27 percent on PPP around 1950. The "anomaly" framing is exactly right.

Japan: about 18 percent to about 4 percent. Nominal. Holds. Peak about 18 percent in 1994 (super-yen + bubble); about 3.7 percent in 2024. On PPP, Japan never exceeded about 8 percent, most of the "fall" is currency unwinding plus slow growth.

Europe (incl. UK): about 30 percent (1992) to about 17 percent. Mixed. Roughly holds, but blended. The "about 17 percent" matches PPP for Europe; nominal EU+UK today is closer to about 21 percent. The "about 30 percent in 1992" is nominal. Two rulers in one sentence.

China about 40 percent before the 1800s. PPP. High-end estimate. Maddison's GDP peak for China is closer to a third (about 33 percent in 1820). The bigger figure is often the manufacturing share, not GDP. Contested on level and timing.

US: 5 percent of population, 25 percent of GDP. Both. Holds. Multiplier about 6 nominal, about 3.6 PPP. The US is about 4.2 percent of world population, so the per-capita lead is, if anything, understated by "5 percent."

India left out of the story. PPP & multiplier. The omission. India was about 24 percent of world GDP in 1700, fell to about 3 percent nominal, and is back to about 8 percent PPP, yet at a multiplier of just about 0.46, a stage behind China, and the largest single swing factor on the board.

GDP share equals power. No single ruler. Incomplete. GDP is capacity; power requires conversion through finance, technology, institutions, military logistics, energy, and alliances.

## Section 18. Conclusions

Seven things the numbers actually say.

1. Size is headcount; weight is productivity. The multiplier dissolves the myth of a "rich" pre-industrial Asia: China's 33 percent of 1820 was a 0.9 times multiplier, many people at subsistence, not a wealthy society. Aggregate size has always been mostly a population story.
2. The ruler must match the question. PPP for living standards, nominal for financial and geopolitical weight. Neither is "true"; the gap between them is information, and the choice of ruler is itself a move on the board.
3. The US didn't decline, its allies did. America held its about 25 percent nominal share for four decades; Europe and Japan ceded the ground that went to China. But allied decline is a double edge: it raises relative standing while eroding the coalition that converts standing into power.
4. China's rise is real in output, barely begun in prosperity. It is the largest economy on PPP and has lifted more people from poverty than any society in history, yet it only just crossed the world-average multiplier, and now faces the middle-income fork: glide to the frontier, or stall at the institutional ceiling.
5. The nominal/PPP gap is a clock. By Balassa, Samuelson it closes from below as a converging economy's prices rise, and stalls if productivity growth stops. China's nominal share will drift up toward its PPP share, unless the clock stops.
6. India is the wildcard; Africa is the next omission. With 18 percent of humanity and a 0.46 times multiplier, India's modest per-head gains move the whole pie. And the "Rest of World", already over a third of output, Africa above all, is where the missing giant of 2050 is being born.
7. The frontier decides the future. Every path turns on productivity, and productivity now means energy and AI. The next rebalancing will be settled by knowledge and power, not headcount, and it should be read with humility, because the rulers themselves are wobbly to ±5 to 10 percent.

The final answer is not "America is declining" or "China is replacing America." It is sharper: America remains a high-multiplier, high-nominal-weight economy; China

has become the largest real-output aggregate in PPP terms but still sits near the world average per person; India is the biggest under-discussed swing factor; Europe and Japan are rich but smaller in relative share; and the next era will be decided by technology diffusion, institutions, energy, demographics, and the political ability to turn output into power. Which side of the world average the largest society on earth comes to rest on, and who, in the next decade, gets to decide, no ruler can tell you. Only its people can, with whatever they are allowed to make of their days.

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## The Long View. Scenarios and Implications to 2050

A speculative extension, more exploratory than the essay above, and held to a lower evidentiary bar on purpose. It was developed through three panels run in designed disagreement, five rounds each: a deep-thinkers roundtable for the structural and civilizational analysis (with Ibn Khaldun, Sun Tzu, Rachel Carson, and Hannah Arendt called in), an fx-experts assessment for the monetary-order trajectory (its Turkish-lira brief adapted to the global reserve-currency question), and a wisepersons panel for the epistemic and ethical frame (Haraway, Morrison, Kahle, adapted from their native domain). What follows is the synthesis; the dialogue is not printed. Read it for shape and stakes, not as forecast, the rulers are wobbly to ±5 to 10 percent, the horizon is far, and the discontinuities at the end of this part would overwrite any baseline drawn before them.

The essay proper ended at a wall it would not climb: which way the slices move next, no ruler can tell you. This part climbs the wall anyway, with the humility the height demands. It does not predict 2050. It maps the space of credible 2050s, names the forces that select among them, and asks the questions the bare shares cannot, for whom each future is good, who gets to author it, and what survives if the system it assumes breaks.

### The Four Drivers, And the Fifth That Cuts Across

Twenty-four years is not long. Much of 2050 is already determined: the workers of 2050 are alive now, the capital stock turns over slowly, and institutions change at

the pace of generations. So the panels began by separating what is nearly fixed from what is genuinely open, and four drivers survived as the levers that will select which future arrives.

Productivity, the AI-and-energy question. This is the only term that compounds, and the one true wildcard. The next general-purpose technology, artificial intelligence layered on cheap electricity, will either raise the frontier where capital, compute, and talent already cluster, or diffuse cheaply into every phone and clinic. Nothing else on this list moves the multiplier as far or as fast.

The monetary order, the dollar question. Whether the dollar-centric system holds, erodes gently, or fragments into rival currency spheres governs the nominal ruler, financial command, the ability to borrow, sanction, and settle, independent of who produces the most real output. A country can lead in PPP volume and still be locked out of the systems only dollars reach.

Demography, the most predictable driver. The working-age wave is already cast. India and sub-Saharan Africa carry a young, growing labour force; China, Europe, Japan, and Korea are aging and beginning to shrink. Demography rarely decides the winner, but it sets the tailwinds and headwinds everyone else must work against.

Institutional vitality, the Ibn Khaldun driver. The least visible and, the panel's sharpest voice argued, the most decisive over decades. The middle-income trap, elite capture, fiscal indiscipline, the slow ossification of the institutions that built a boom, these are what stall a converging economy or hollow a frontier one. The deepest risk to both the United States and China is internal, not the rise of the other.

Figure 20. Demographic destiny: working-age (15 to 64) population, indexed to 2024 = 100, projected to 2050. Diverging lines: sub-Saharan Africa and India climbing steeply, the United States roughly flat, China and Europe declining, Japan falling fastest. The one driver the chart can state with near-certainty, because its 2050 cohort is already born.

Cutting across all four is energy and ecology, not a side-constraint but a primary force. The cost of decarbonised power and the politics of critical minerals will reshape where things are made; a serious climate shock or a minerals-supply rupture is a live tail. Cheap clean electricity lowers the cost of making everything and is the substrate AI runs on; expensive or contested energy raises every other

cost and splits supply chains. Energy is the medium in which the other four drivers act.

### The Scenario Space

Two of the four drivers are both high-impact and genuinely uncertain, the frontier (does AI-and-energy concentrate productivity or diffuse it?) and the order (does the global system integrate or fragment?). Crossed, they generate four coherent worlds. A fifth, the stagnation baseline, is orthogonal: it is what any of the four becomes if productivity simply disappoints. Demography and institutions then determine who occupies which position within each world.

Figure 17. The scenario space. A 2 times 2: the horizontal axis runs from the frontier concentrating (left) to diffusing (right); the vertical axis from the order fragmenting (bottom) to integrating (top). The four quadrants are named below; a fifth scenario, the Long Plateau, sits across the centre as the low-productivity version of all four.

Frontier Pull-Away (integrated order, concentrated frontier). One world market, but AI-and-energy gains compound where they already cluster, the United States, China, a handful of hubs. The frontier economies' multipliers widen; the catch-up stalls; inequality between nations grows inside an integrated system. This is §14's "frontier discontinuity," refined: re-divergence without fragmentation. The US re-anchors on a productivity boom and the dollar with it; China holds its scale but the gap to the frontier stops closing.

The Great Catch-Up (integrated order, diffused frontier). The optimistic path. Cheap intelligence and cheap power diffuse fast, a tutor, a clinician, an engineer in every phone, and become the largest catch-up accelerant ever built. India's multiplier climbs toward and past the world average; Africa industrialises a generation faster than the historical script; the gap between nations narrows for the first time since the Great Divergence. World growth is high; the shares converge toward population weight. The world of 2050 is multipolar and richer.

Two Suns (fragmented order, concentrated frontier). The most dangerous world. The system splits into a US-led and a China-led techno-economic sphere, each with its own standards, payment rails, supply chains, and currency bloc; the frontier concentrates within each sphere; the unaligned, India above all, much of the Global

South, hedge between them and extract rents from the rivalry. "World GDP share" stops being one statistic: there are two pies, measured in rival units, and the nominal ruler itself bifurcates.

Archipelago (fragmented order, diffused frontier). Technology diffuses, but the order fragments into several regional spheres rather than two camps. More autonomous regional economies; diffusion lifts many, but the gains from a single integrated market are lost to friction. Messy, plural multipolarity, neither the clean rivalry of Two Suns nor the open integration of the Catch-Up.

The Long Plateau (any order, productivity disappoints). The baseline the panels kept returning to as the modal near-term reality. AI underdelivers on measured growth; aging, debt overhangs, and middle-income traps bind; no one pulls away and no one catches up fast. Shares drift slowly, almost entirely by demography. A low-growth, multipolar, integrated-but-sluggish world, secular stagnation as the global default. Not a catastrophe; a long, grey flatness.

Figure 18. The cone of outcomes. World GDP share at PPP for the United States, China, and India, 2024 to 2050, each drawn as a shaded fan spanning the five scenarios. The fans widen with time and overlap: China's upper edge (Pull-Away/Two Suns scale) and India's upper edge (Great Catch-Up) are the widest, the United States' the narrowest, a visual statement that the most uncertain economies are the populous risers, not the incumbent.

### What Each Scenario Does to the Map

The scenarios are not equally kind to each actor, and the same economy can be a winner in one and a casualty in another. The table reads the five worlds as rough 2050 PPP-share directions, not point forecasts, the arrows are the message.

United States. ↑ ↑ re-anchors on AI productivity; dollar strong. ↓ relative, as others rise; richer absolutely. ↑ leads one sphere; financial command intact. ↓ one pole among several. to drifts, aging-light, holds quarter slowly

China. to holds scale, gap to frontier stalls. ↑ converges, but aging caps it. ↑ ↑ anchors the rival sphere; most leverage. ↑ regional hegemon of its zone. to demography and debt bite; slow plateau

India. ↓ left behind the frontier. ↑↑ the largest single gain on the board. ↑ the swing prize both spheres court. ↑ leads its own regional zone. ↑ slowly; demography carries it alone

Sub-Saharan Africa. ↓↓ the omission deepens. ↑↑ industrialises a generation early. to courted, fragmented, hedging. ↑ regional integration helps. to population grows, output lags

EU + UK. ↓ falls further behind the frontier. to holds, modernises at the margin. ↓ squeezed between two spheres. to a regional pole, diminished. ↓ aging and stagnation compound

Rest of World. ↓ frontier rents bypass them. ↑ broad diffusion lifts the floor. to bloc-aligned or hedging. ↑ regionalism favours them. to flat

Figure 19. 2050 PPP shares under each scenario. Grouped bars: the United States, China, India, sub-Saharan Africa, EU+UK, and Rest of World, shown five times across the scenarios. The eye should catch two things, how far India's and Africa's bars swing between Pull-Away and Catch-Up, and how little the United States' bar moves regardless, the incumbent's share being the most insured against the future.

### The Monetary Order to 2050

The fx-experts panel, asked to look past the lira to the system itself, returned a house view notable for its restraint: the dollar order most likely persists but erodes, rather than either enduring untouched or collapsing. The dollar's reserve share, having drifted from roughly 70 percent in 2000 to about 58 percent now, call it half a point a year, most plausibly reaches the mid-to-high 40s or low 50s by 2050: a more multipolar system in which the dollar remains primary because no rival combines the three things reserve status requires at once, a deep market of safe assets, an open capital account, and the rule of law, at sufficient scale. The euro lacks the unified fiscal backing; the yuan is structurally capped below 3 percent precisely because opening the capital account would mean surrendering the capital controls the Chinese state will not cede. The reserve role is far stickier than the de-dollarisation headlines, which mostly track bilateral trade settlement, small relative to reserves and capital markets, rather than the reserve and funding roles that actually confer power.

Figure 21. The dollar's reserve share to 2050, four paths. From about 58 percent in 2024: a status-quo-plus line (an AI-driven US productivity boom and safe-asset demand hold it near 55 percent), a base drift (toward about 48 percent), a fragmentation path (a rival bloc forms; toward about 40 percent), and a low-probability rupture (a disorderly fiscal-crisis repricing; a faster, steeper fall). The reserve premium can erode for decades and then break in months, the lines fan slowly, but the rupture tail is drawn for a reason.

The asymmetry the panel kept returning to is the one the essay flagged: the dollar is a crisis hedge. In a global shock, capital flees toward it, so the US nominal share tends to rise exactly when the world is most frightened. Betting against the dollar order is, in effect, shorting a convexity that pays off in tail events, which is why the order is more durable than its critics expect and more fragile than its defenders admit. The fattening tail is fragmentation as policy: sanctions, export controls, and reserve freezes are actively pushing the system toward bipolarity from both sides, and a hardening into a rival settlement bloc, gold, the yuan in trade, CBDC rails for the sanctioned and the hedging, is what would move the dollar's share fastest and split the nominal ruler into two. The minority risk: a disorderly dollar repricing from twin deficits and a politicised central bank is low-probability but under-priced, and it would dominate the nominal picture overnight.

### The Frontier Question, The One Variable That Matters Most

If only one driver could be known, the panels agreed it should be this: does the AI-and-energy frontier concentrate or diffuse? Because it acts directly on the only term that compounds, it has the widest fan of any single variable, wider than demography, wider than the dollar. The same technology runs two opposite vectors. Concentrated, it is the greatest re-divergence engine since coal and steam: gains compound where data, capital, and elite talent already sit, the frontier multipliers widen, and the Long Divergence gets a second act. Diffused, it is the greatest convergence engine ever built: the marginal cost of expertise collapses, and a clinic in Patna or Lagos gets a diagnostician, a tutor, an engineer it could never staff.

Figure 22. Diffuse versus concentrate, drawn on the multiplier. Projected change in PPP output per head relative to the world average, 2024 to 2050, for frontier economies (United States, EU) against catch-up economies (India, sub-Saharan

Africa), under the two AI vectors. Under concentrate, the frontier lines rise and the catch-up lines flatten, the gap widens. Under diffuse, the catch-up lines rise sharply and the frontier lines plateau, the gap narrows. Same technology; opposite worlds; one chart.

Which vector wins is not purely technological. It is set by whether the gains are captured or shared, by competition policy, by who owns the compute and the models, by whether cheap power reaches the places that need it, and by the institutions that turn a tool into broad productivity. Chips without electricity, models without institutions, and intelligence without the schooling to use it do not lift a multiplier. The frontier question is, underneath, a political question wearing a technological costume.

### The Discontinuities, What Overwrites the Baseline

The scenarios above are slow-driver baselines. The panel's most insistent minority view, and the wisepersons' resilience reading, is that the tails carry more expected impact than the modal path, because any one of them rewrites the board faster than demography or institutions ever could. A forecast that names only the smooth paths is a forecast lying about its own confidence.

The fat tails worth naming: a war over Taiwan, which would fracture the semiconductor frontier and the trade order at once and force the Two Suns world overnight; a disorderly dollar repricing from a US fiscal crisis or a politicised Fed; an AI discontinuity, a genuine jump to broadly transformative systems, that concentrates power faster than any institution can redistribute it, or, inversely, an AI disappointment that collapses the valuations and the growth premium the optimistic paths assume; a pandemic worse than the last; and a climate or minerals shock, a breadbasket failure, a critical-minerals chokepoint, that raises the cost of everything and splits supply chains. Each is low-probability in any given year and high-probability across a quarter-century; each would dominate the slow drivers if it landed. The honest posture is Tao's: the future of economic weight is a fat-tailed, asymmetric distribution, and the baseline is mostly a story we tell to fill the space the tails will likely overwrite.

### The Standpoint and the Stakes

The wisepersons panel refused to let the section end as a power-ranking, and its objection is part of the finding rather than a footnote to it. Three corrections to everything above.

Whose future is this, and who authored it? Every scenario here is structured around the powers that already hold weight, the United States, China, India, the EU, the dollar. That is a view from the centre, written from the vantage of those who measure weight, and the vantage shapes which futures get imagined. The clearest symptom is the one the essay already named as Kotkin's error and then half-repeated: sub-Saharan Africa, which will hold roughly one in four human beings by 2050 and supply most of the planet's population growth, appears in the standard scenario literature, and nearly appeared here, as a "swing factor" and a "Rest of World" band rather than as a set of societies with futures of their own authorship. The next missing giant is being born there now, and it does not get a vote in the scenarios that describe its life. Naming this is not a caveat to the analysis; it is a correction to it.

The decimal hides a person, at the horizon too. "India's multiplier climbs to 0.7" and "Africa industrialises early" are abstractions for whether several hundred million people cross from precarity to security, whether a child gets a grid, a clinic, a school. The Great Catch-Up is not, at bottom, a redistribution of share-points; it is the largest improvement in human lives the century could contain. The Long Plateau is not a grey flatness on a chart; it is a generation in the Global South that does not get the convergence the last one was promised. The scenarios are worth ranking by that, not only by who leads.

Efficiency is not resilience. The macro panels optimise for who will be biggest; the preservationist's question is what survives when the system breaks. A hyper-optimised, concentrated, just-in-time global economy, the Frontier Pull-Away world, is also the most fragile to any of the discontinuities above. The resilient futures are the redundant, somewhat-localised, plural ones that trade efficiency for survivability. That efficiency-versus-resilience tradeoff is a dimension the share-charts cannot show and the next decade will keep testing, and it is, quietly, an argument that the messier Archipelago may be safer than the cleaner worlds on either side of it.

## What to Watch

The leading indicators that tell you which way the drivers are turning, the dials to read over the next decade, organised by the driver each one tracks.

- Productivity / AI. Measured productivity-growth data in the US and China; whether AI gains show up in output per hour broadly or only in a few firms; compute and electricity build-out. Broad productivity gains to diffusion and Catch-Up; narrow gains to concentration and Pull-Away
- The monetary order. Dollar reserve share (annual COFER); the gap between the yuan's trade-settlement share and its reserve share; US debt/GDP and term-premium behaviour; non-dollar rails (CIPS, mBridge, CBDCs); commodity invoicing units. A closing yuan reserve gap or commodity de-dollarisation to fragmentation; a stable gap to slow drift
- Demography. India's labour-force participation, especially female; African industrial job creation; China's and Europe's old-age dependency ratios. Participation rising where the workers are to tailwinds convert to output
- Institutions. China's response to the middle-income fork; US fiscal trajectory and institutional cohesion; reform versus capture across the Global South. Reform to convergence; capture and indiscipline to stall, on the Ibn Khaldun pattern
- Energy / ecology. The falling cost of clean electricity; critical-minerals concentration and chokepoints; climate-shock frequency. Cheap diffuse power to Catch-Up substrate; contested or shocked energy to fragmentation and the tails
- The swing variable. India's GDP per head crossing ~$5,000; sub-Saharan Africa's manufacturing share beginning to rise. The single largest latent move on the board beginning to cash

## Closing the Long View

Strip the five worlds and the six dials down, and the same hinge that ran through the essay runs through its future: productivity, and specifically whoever commands the energy-and-knowledge frontier, and whether they hoard it or share it. Population set the stage and the ruler set the framing, but the third term decides which 2050 we wake up in. The modal near-term path is a muddle, the frontier pulling away somewhat while growth stays mediocre, with fragmentation a rising probability and

the tails fattening underneath. The most leveraged unknown is the AI vector; the most decisive slow risk is internal, legitimacy and discipline, in Washington and Beijing alike; and the most under-counted actors are the African societies the maps keep drawing as a residual.

And the part the rulers will never measure stays exactly where the essay left it. Whether the next quarter-century is the Great Catch-Up or the Long Plateau is not, finally, a fact waiting in the data. It is something hundreds of millions of people will make, or be prevented from making, out of whatever they are given and whatever they are allowed. The charts can show you the shape of the room. They cannot tell you what the people in it will choose to build, only that, this time, far more of them will be in a position to choose.

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## Appendix A. Data & Methodology

Three source lanes, kept deliberately un-stitched.

This report uses several source lanes and keeps them separate by design. The long-run historical view is Maddison-style; the modern 1990 to 2024 view is World Bank WDI variables (as carried through the QoG standard country-year file); the present-day marker is the IMF WEO April 2026 vintage. They are related, but they are not one seamless instrument.

The core calculations are simple. GDP share is entity GDP divided by world-country-sum GDP under the chosen ruler. Population share is entity population divided by world-country-sum population. The multiplier is GDP share divided by population share, 1.0 means world-average output per person; 4.0 means roughly four times the world average.

> Maddison Project Database (Bolt & van Zanden, 2023 update). 1500 to 2008 benchmark shares and long-run multipliers; folds in Bassino & van der Eng and Ma & de Jong East-Asia studies.. Historical borders, PPP benchmarks, sparse data, reconstruction uncertainty. The workbook effectively ends about 2008.
>
> World Bank WDI / QoG modern country-year file. 1990 to 2024 nominal shares, PPP shares, population shares, and multipliers.. Country-sum denominators and

variable definitions can differ from official IMF/global aggregates by a few tenths of a point.

IMF WEO April 2026 marker. Current projection marker for nominal GDP, PPP GDP, PPP shares, and macro aggregates.. A projection vintage; not spliced backward into historical series.

UN World Population Prospects 2024. World population and projections to 2050/2100, anchoring the multiplier denominator.. Medium-variant demographic projection; not an economic forecast.

Bairoch-style manufacturing series. Industrial-power context during the Great Divergence.. Approximate historical manufacturing shares; used for direction, not decimals.

Why not stitch everything into one continuous 1500 to 2026 line? Because the appearance of continuity would be false precision. Long-run Maddison reconstructions, WDI modern PPP series, and IMF projection markers use different vintages, benchmark systems, and assumptions. A visually smooth line would look more scientific while being less honest. The deeper into the past a chart reaches, the more its decimal point is a costume.

On the three PPP totals. World PPP GDP appears in three forms in this report, deliberately distinguished: $222.8 trillion (IMF April 2026, 2026 projection, current international dollars, the present-day marker); ~$195 trillion (2024, current international dollars, the modern snapshot); and ~$170 trillion (2024, WDI constant 2021 international dollars, the internal real-output arithmetic). They are not interchangeable, and the figures are not averaged across lanes.

Uncertainty discipline. Pre-1900 numbers are reconstructed estimates with wide error bars. Modern numbers are more precise but still depend on ruler, denominator, source vintage, and grouping. PPP itself carries ±5 to 10 percent structural uncertainty and has lurched by tens of percent for China across ICP rounds. Future numbers are scenarios, not forecast certainties.

## Appendix B. The Reference Tables

The numbers under the charts, gathered in one place so the figures can be reproduced.

B1, World GDP, two rulers and two multipliers, 2024 (WDI country-sum). Shares against a world total of roughly $110 trillion nominal / ~$170 trillion PPP (constant 2021 int\$, about $195 trillion current); they differ by a few tenths from aggregations built on official world totals. EU-27 + UK is a constructed sum. The decimals are calculation artifacts, read the ruler, the order of magnitude, and the gap.

United States: about 26 percent of world output at market rates, about 15 percent on the PPP ruler, and 4.23 percent of world population. Its output-per-person multiplier is 6.23 at market rates and 3.56 on PPP.

China: about 17 percent of world output at market rates, about 19 percent on the PPP ruler, and 17.51 percent of world population. Its output-per-person multiplier is 0.98 at market rates and 1.13 on PPP.

EU-27: about 17 percent of world output at market rates, about 14 percent on the PPP ruler, and 5.59 percent of world population. Its output-per-person multiplier is 3.20 at market rates and 2.57 on PPP.

EU-27 + UK: about 21 percent of world output at market rates, about 16 percent on the PPP ruler, and 6.45 percent of world population. Its output-per-person multiplier is 3.29 at market rates and 2.56 on PPP.

United Kingdom: about 3 percent of world output at market rates, about 2 percent on the PPP ruler, and 0.86 percent of world population. Its output-per-person multiplier is 3.93 at market rates and 2.48 on PPP.

India: about 3 percent of world output at market rates, about 8 percent on the PPP ruler, and 18.03 percent of world population. Its output-per-person multiplier is 0.20 at market rates and 0.46 on PPP.

Japan: about 3 percent of world output at market rates, about 3 percent on the PPP ruler, and 1.54 percent of world population. Its output-per-person multiplier is 2.40 at market rates and 2.18 on PPP.

Russia: about 1 percent of world output at market rates, about 3 percent on the PPP ruler, and 1.78 percent of world population. Its output-per-person multiplier is 1.10 at market rates and 2.00 on PPP.

B2, The IMF April 2026 current marker. Kept on its own page and deliberately not stitched into the long-run series (different vintage, different benchmark system). World totals: nominal $126.3 trillion; PPP $222.8 trillion, both in current dollars / current international dollars. Nominal shares computed against the $126.3 trillion world total; EU + UK is a constructed sum.

United States. 14.54. 32.38. 25.64

China. 19.89. 20.85. 16.51

India. 8.49. 4.15. 3.29

European Union. 13.77. 23.03. 18.23

EU + UK. 15.89. 27.30. 21.62

United Kingdom. 2.12. 4.26. 3.37

Japan. 3.26. 4.38. 3.47

Russia. 3.38. 2.66. 2.11

How to use the marker: the IMF row is a current projection snapshot, world prices, current exchange rates, current international-dollar PPP. The WDI/QoG country-sum table (B1) carries the report's internal 2024 share and multiplier arithmetic. Maddison-style data carries the centuries. The lanes touch, but they are not the same road.

B3, The centuries, in one table (Maddison-style, 1990 international dollars).

1500. about 0.44 billion. $0.25 trillion. ~$565. Agrarian scale: output mostly follows people, land, calories.

1700. about 0.60 billion. $0.37 trillion. ~$615. Still a population-weighted world; China and India remain huge in aggregate.

1820. about 1.04 billion. $0.69 trillion. ~$665. Industrial divergence beginning; old agrarian weights still visible.

1870. about 1.27 billion. $1.11 trillion. ~$870. Steam, coal, railways, finance begin to widen the per-person gap.

1913. about 1.79 billion. $2.73 trillion. ~$1,525. Before WWI, industrial output has remade the economic map.

1950. about 2.53 billion. $5.34 trillion. ~$2,110. The postwar American peak sits inside a world still rebuilding.

1973. about 3.92 billion. $16.02 trillion. ~$4,090. Postwar growth multiplies world output faster than population.

2000. about 6.15 billion. $36.69 trillion. ~$5,970. Globalisation raises scale; China and India re-enter modern production.

2008. about 6.71 billion. $50.97 trillion. ~$7,600. End of the Maddison long-run GDP lane used here.

Modern totals (2024: ~$110 trillion nominal current US\$; ~$195 trillion PPP current int\$; ~$170 trillion PPP constant-2021 int\$) use different statistical lanes and are not spliced onto this scale.

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## Appendix C. Bibliography & Data-Source Urls

Sources used for definitions, data lanes, and the inspiration reference. The data essay avoids full academic footnoting in the body to stay readable; this appendix and the build bundle are the audit trail.


Hoover inspiration. youtube.com/watch?v=gBEdxb8ei_0. User-specified source: Uncommon Knowledge, Hoover Institution; guest Stephen Kotkin, host Peter Robinson.

Hoover episode page. hoover.org/research (Kotkin / Uncommon Knowledge). Public episode identification; host/guest, date, title context.

World Bank ICP methodology. worldbank.org/en/programs/icp/methodology. PPP concept, price-level equalisation, comparison logic.

WDI PPP GDP metadata. databank.worldbank.org, series NY.GDP.MKTP.PP.KD. Definition of PPP GDP in constant 2021 international dollars.

WDI nominal GDP. data.worldbank.org/indicator/NY.GDP.MKTP.CD. Official WDI world current-dollar GDP series through 2024 ($111.3 trillion, 2024).

WDI population metadata. databank.worldbank.org, series SP.POP.TOTL. Population denominator: de facto midyear estimates.

Maddison Project Database 2023. rug.nl/ggdc/historicaldevelopment/maddison. Long-run comparative economic history and benchmark context.

IMF WEO April 2026. imf.org/en/publications/weo/issues/2026/04/14. Current macro/projection marker ("Global Economy in the Shadow of War").

IMF DataMapper, PPPSH, NGDPD, PPPGDP, LP. imf.org/external/datamapper (@WEO). Current PPP-share, nominal GDP, PPP GDP, and population markers; world nominal $126.3 trillion, world PPP $222.8 trillion (2026).

UN World Population Prospects 2024. population.un.org/wpp. World population estimates and projections to 2100 (2024, 2050, peak, 2100).

QoG Standard dataset. gu.se/en/quality-government/qog-data. Convenient modern country-year file carrying WDI variables.

Bairoch (manufacturing). "International Industrialization Levels from 1750 to 1980". Manufacturing-output shares during the Great Divergence.

Scholarly context used in interpretation includes the Great Divergence debate (Pomeranz; Broadberry et al.), industrialisation and manufacturing-share histories, the middle-income transition literature, the Balassa, Samuelson framework, and Frankel's rule on matching ruler to question.

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## Appendix D. Colophon

A short account of the instrument, because the essay spends so long insisting the instrument matters.

Edition. Eighth edition, June 2026, the first living edition, built on the seventh-edition fusion. The seventh edition built on the sixth Claude data-essay edition as its argumentative spine and ingested the reporting apparatus (denominator section, the full PPP limitations chapter, the source-lane methodology, the bibliography, and the provenance trail) from the parallel GPT-5.5 Pro publication edition. It is not a symmetric merge: the spine, prose, and original analysis are the sixth edition's; the audit layer is imported where it strengthens defensibility, and the GPT draft's

scaffolding bloat and redundant tables were deliberately not carried over. This build additionally adds an up-front global-totals panel ("The World This Essay Divides") and an extended speculative coda, The Long View: Scenarios and Implications to 2050, developed through three designed-disagreement panels, lifting the figure count from 16 to 22.

What the seventh-edition fusion corrected. The sixth edition's world-PPP figure (~$205 trillion) was an error. Verified directly against IMF WEO April 2026, world PPP GDP for the 2026 projection is $222.8 trillion in current international dollars (the US is the PPP numeraire, so its PPP GDP equals its nominal $32.38 trillion; at the IMF's 14.5 percent US PPP share, the world total is $222.8 trillion). The three PPP lanes, 2026 current ($222.8 trillion), 2024 current (~$195 trillion), 2024 constant-2021 (~$170 trillion), are now explicitly distinguished and never spliced. World nominal ($126.3 trillion), World Bank 2024 nominal ($111.3 trillion), and the IMF country figures all check out against source.

What the eighth edition added. This edition is the first to live in a public, version-controlled repository rather than being rebuilt by hand from chat to chat. Three things were formalised. A machine-readable consistency suite (fifty-two checks across the identity arithmetic, the source-lane separations, and every cross-figure relationship) now runs against the data file before any build. A standing review body, nine experts and fifteen on-call advisors, replaces the ad-hoc panels of earlier editions. And a four-pass finalisation (logic and mathematics; figures between and within; a literary pass; an editorial pass) was run against the whole document. That finalisation corrected four figures in the speculative coda: the cone of outcomes was redrawn so that its 2050 fan is exactly the envelope of the five scenario bars; each scenario's shares were made to sum to one hundred percent with a residual "rest of world"; Japan's price-adjusted line was trimmed so it honours the prose claim that it never crosses about eight percent; and one deep-history population figure was aligned to its own table. No analytical prose changed, because the coda speaks in directions, not point forecasts, which is the property that lets the numbers refresh without the argument moving.

Direction & authorship. Directed by Hulki Okan Tabak; written by Claude 4.8 and Hulki Okan Tabak.

Models. The sixth-edition spine was composed with Claude (Anthropic); the imported apparatus originated with GPT-5.5 Pro. This fused edition was assembled and reconciled on Claude Opus 4.8, which verified the IMF figures against source and ran the panels below.

Inspiration. A public conversation on the Hoover Institution's Uncommon Knowledge, guest Stephen Kotkin, host Peter Robinson, at the link the director supplied. This essay is an independent analysis built from that stimulus; it is not a transcript, an endorsement, or a Hoover publication, and it attributes no specific figure to the conversation.

Panel method. Each panel ran as designed disagreement, not theatre, and the chair does not vote. For the essay core: a deep-thinkers roundtable (logical integrity, hidden assumptions) and an editorial-pass under the fifteen criteria (consistency, logic, typography, front/back matter, no-decimal-exceeds-its-source). For the speculative coda, The Long View: three panels at five rounds each, deep-thinkers (the structural and civilizational analysis, with Ibn Khaldun, Sun Tzu, Rachel Carson, and Hannah Arendt called in), fx-experts (the monetary-order trajectory, its Turkish-lira brief adapted to the global reserve-currency question), and a wisepersons panel (the epistemic and ethical frame, Haraway, Morrison, and Kahle, adapted from their native AI-Skills-governance domain to supply standpoint, the African absence, the human-stakes reading, and the resilience-versus-efficiency axis). The author ratifies; the panels argue; the surviving synthesis is incorporated; the dialogue is not printed.

Build. One source of truth. The same numbers and prose render three ways, an interactive Chart.js page, a typeset PDF, and a screen-reader edition, with every figure drawn from the data file, never hand-placed. Change a number once and all editions follow. From the eighth edition onward this pipeline lives in a public repository and rebuilds itself when the source data updates; the mechanism is described in Appendix F.

Typography. Fraunces for display and italics, IBM Plex Sans for text, IBM Plex Mono for labels. Oxblood (#993C1D) on warm paper.

Discipline. PPP carries ±5 to 10 percent structural uncertainty; deep-history figures are reconstructions with wide error bars. Read the ruler first, the direction second, and the decimal point last.

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## Appendix E. The Project in Numbers

What went into this edition. Recorded because a thinking aid should be auditable.

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## Appendix F. This Is a Living Document

Why the eighth edition has a version number and a repository, and what keeps it honest as the numbers move.

Every edition before this one was a static object: a manuscript and a build pipeline that a person carried, by hand, from one conversation to the next, rebuilding the figures each time a number changed. That worked, but it made the essay mortal, dependent on a single chat's memory, and out of date the moment the IMF or the World Bank published a revision. The eighth edition is built differently. Its single source of truth, one data file of numbers, one prose manuscript, and the renderers that turn them into the three editions you can read, now lives in a public, version-controlled repository on GitHub, one that anyone may read, copy, or fork. The document you are holding is a snapshot of that repository at a fixed version; the repository itself is designed to update, and the ordinary way to propose a change, a corrected figure, a newly released number, an improvement to the build, is to fork it and open a request to merge. Updating it is an option, not an obligation: an open invitation to anyone who finds the essay useful, and a capability the project keeps available rather than a maintenance promise it makes. You can read the living version online at hulkiokantabak.github.io/weight-of-nations, and the repository, the source of truth behind every edition, is at github.com/hulkiokantabak/weight-of-nations.

The update is deliberately narrow, and that narrowness is the whole trick. The argument of this essay is not a forecast. It is a method, separate the rulers, divide out population, watch productivity, and a set of directions: who is rising, who is

plateauing, which gap is closing. Methods and directions do not expire when a denominator is revised from $205 trillion to $222.8 trillion; only the decimals do. So the living mechanism touches only the data file. When the IMF's October World Economic Outlook lands, or the World Bank refreshes its indicators, or the UN issues a new population revision, the numbers are updated in one place, the consistency suite re-runs its fifty-two checks, and all three editions rebuild from the corrected source. The prose is not rewritten. If a number moves far enough that a direction changes, if a plateau becomes a decline, or a laggard begins to converge, that is no longer a data refresh but a new analytical edition, and it goes back through the review body before it ships.

The architecture is dual. Two instances of the assistant do different jobs, and keeping them separate is what keeps the document disciplined. A chat advisor plans, reads the figures as a critic would, ratifies changes against the project's standing rules, and prepares the exact instructions for a change, but never edits the manuscript or the data directly. A code executor receives those instructions, makes the single change, re-runs the checks, rebuilds, and commits the result to the repository with a message that records what moved and why. The advisor thinks; the executor builds; neither does the other's job. This is the same division of labour that produced the essay's earlier editions, now written down as the operating rule of the living document.

The analysis has a standing review body. Earlier editions convened panels ad hoc, a roundtable here, an editorial pass there. The eighth edition formalises them into nine resident experts, each holding one lens the others do not: two economists, one on growth and convergence and one on measurement and the rulers; an economic anthropologist who keeps the lived body in view behind the aggregates; two technologists, one on the diffusion of the frontier and one on its physical substrate of compute and power; a specialist in institutions and legitimacy; two historians, one of the long divergence and one of the postwar order; and a demographer who also holds the geopolitical fan, the most certain variable and the least certain in one seat. Fifteen on-call advisors are summoned for the narrower questions a given edition raises, from semiconductors and energy transition to sovereign debt, migration, trade routes, and data quality. They argue by design; the chair

synthesises but does not vote; the author ratifies. The panel's full composition and triggers live with the project's skills.

The instruments behind the document are themselves recorded, because an essay that spends fifteen thousand words insisting the instrument matters should disclose its own. The repository carries the manuscript, the data file, the renderers, the consistency suite, a methodology that explains how each edition was made, a running log of every error caught and the standing rule it produced, a metrics file that counts the editions and the checks, and a small set of skills, one that knows how to update the document and rebuild it, one that convenes the review body, and one that codifies the annual data-refresh ritual and the source for every number. Together they are the audit trail. The point of all of it is the same as the point of the essay: a number is a costume, a direction is the body underneath, and the only way to keep the two from being confused is to show your work, and to keep showing it as the work changes.

The shape of the project. The living document is one of three layers, and naming them keeps the claim honest. There is a fixed, dated edition, this one, for readers to quote; a living public repository that holds the data, the code, the claim registry, and the correction log; and a skill that codifies the reasoning and is the interface to the repository rather than a second source of truth. A reader meets the project as a book or a web page; a researcher meets it as a repository. The "living" claim is deliberately modest: the editions are dated snapshots, the repository may be updated by the author or by contributors, and it is promised to no one, if it is never touched again, this dated edition stands on its own. That modesty is the same discipline the essay asks of its numbers: say what a thing is, date it, and do not claim more currency than you can keep.